Deepak Mallampati

APPLIED AI / MLOPS ENGINEER

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PROFESSIONAL SUMMARY

Applied AI engineer with 4+ years building production LLM and ML systems and the evaluation, observability, and automation layers that keep them reliable. Established an LLM evaluation & monitoring framework (RAGAS, BERTScore, custom metrics) with latency/accuracy dashboards, and built fault-tolerant automation infrastructure that cut failed production runs by 40–60%. Comfortable owning the full AI lifecycle — prototyping, integration, evaluation, observability, and production support — across cloud and containerized environments.

CORE COMPETENCIES

Production LLM Systems • Agentic Workflows • Evaluation & Observability • MLOps & CI/CD • Fault-Tolerant Automation • API & Systems Integration • Full AI Lifecycle

AI Platform & MLOps: Docker, Kubernetes, Terraform | CI/CD (Jenkins, Azure DevOps) | MLflow, Weights & Biases | Model monitoring & observability | RAGAS, BERTScore | Scheduled orchestration + retry logic

LLMs & Retrieval: GPT-4o/5, Claude, Gemini, LLaMA | LangChain/LangGraph | Multi-agent systems | RAG (FAISS, Pinecone, Weaviate, pgvector) | Fine-tuning (LoRA, QLoRA, PEFT)

Engineering & Cloud: Python, TypeScript, SQL | FastAPI, React | AWS (Bedrock, SageMaker, S3, Lambda), Azure, GCP | PostgreSQL, Redis, Kafka | Spark/Databricks

PROFESSIONAL EXPERIENCE

Progress Solutions *Jul 2025 - Present | USA*

AI / Machine Learning Engineer

- Engineered high-performance **retrieval-augmented (embedding-based) pipelines** over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and reduce hallucination by **40%** at **92% precision**.
- Designed **privacy-preserving data workflows** for sensitive clinical datasets (k-anonymity, l-diversity, Laplace differential privacy), enabling compliant analytics and model development with zero PII exposure.
- Architected **agentic LLM systems** on a conductor–subagent framework, cutting token consumption **42%** while sustaining task accuracy across multi-step workflows.
- Established a structured **ML evaluation & monitoring framework** (RAGAS, BERTScore, custom metrics) with latency/accuracy dashboards to benchmark iterations and catch regressions before release.
- Built fault-tolerant automation (scheduled orchestration, exponential-backoff retries), reducing failed production runs by **60%**.

Kent State University *Oct 2023 - Jan 2024 | USA*

ML & Gen AI Research Assistant

- Built a **privacy-preserving ML pipeline** on a clinical hospital-readmission dataset; reduced re-identification risk from **3.38% to 0.32%** and eliminated unique-record exposure while retaining **99% of baseline predictive performance** (Random Forest AUC 0.689 vs. 0.696).
- Designed a composite **privacy–utility scoring framework** benchmarking six anonymization configurations and quantifying differential privacy's utility cost across four ε levels — a probabilistic trade-off analysis directly analogous to causal/Bayesian modeling.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100; authored a peer-review-ready **IEEE-format conference paper** documenting methodology and results.

Vanguard
Jan 2024 - Jan 2025 | USA

AI Engineer Intern

- Built **12 APIs and microservices** powering AI capabilities, reducing p95 latency 1.5s → **800ms** and supporting **100,000+ requests daily** across 20+ teams.
- Evaluated GPT-4o, Claude Sonnet, and Gemini, improving task success rates by **27%** on internal evaluation datasets and informing model-selection decisions.

Emerson
Jun 2022 - Apr 2023 | India

ML Trainee

- Developed and trained supervised **regression and classification models** on operational data for equipment-failure prediction, anomaly detection, and capacity forecasting, with end-to-end feature engineering and preprocessing.
- Fine-tuned BERT/RoBERTa for entity extraction and text classification, achieving **89% F1** on custom NER tasks.

PROJECTS

Privacy-Preserving Clinical ML Pipeline

Framework for analyzing sensitive patient data without exposing identities, combining k-anonymity, l-diversity, and differential privacy — quantifying exactly how much privacy each technique buys for how little accuracy cost.

career-ops

Autonomous AI job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision.

EDUCATION

M.S. in Computer Science, *Kent State University, OH* *2025 | GPA 3.70/4.0*